

AI-POWERED INTELLIGENT SALES FORECASTING

MILESTONE 4 – SYSTEM INTEGRATION

TEAM MEMBERS

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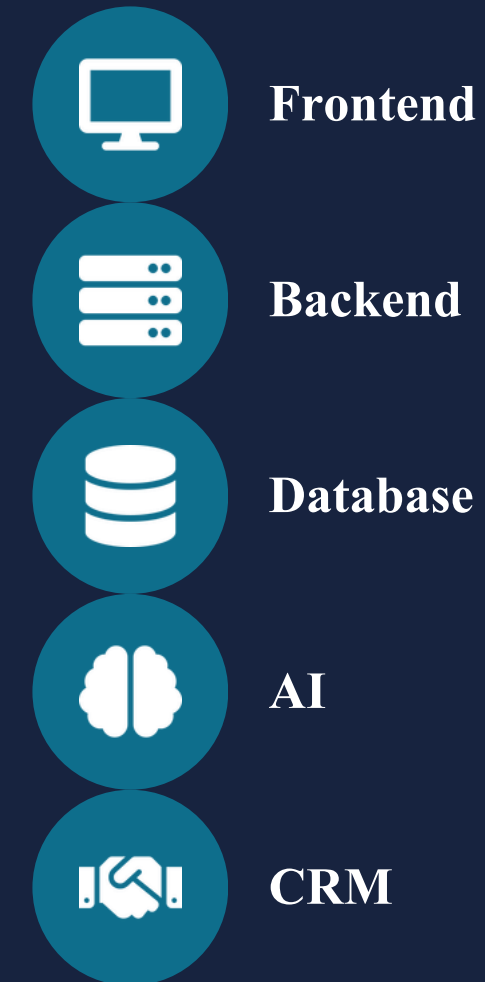
Milestone Focus: Integrating all developed modules into one validated, end-to-end pipeline.



Milestone 4 – Integration

- Integrate the developed modules into a unified application.
- Establish communication between the frontend, backend, database, AI services, and CRM integration layer.
- Enable seamless data flow between different components.
- Validate the integrated end-to-end workflow.
- Measure execution time of major system operations.

COMPONENT INTEGRATION



**INTEGRATED
SYSTEM**



Integration Objectives



Connect All Modules

Unify all implemented modules into a single workflow.



Frontend–Backend Link

Integrate Streamlit frontend with FastAPI backend services.



Database Connectivity

Connect backend services with the PostgreSQL database.



AI Company Analysis

Integrate AI-based company analysis and lead intelligence.



Lead Scoring

Integrate lead scoring and conversion probability prediction.



AI Email Outreach

Integrate AI-powered email outreach generation.



CRM API Integration

Enable CRM-related API integration.

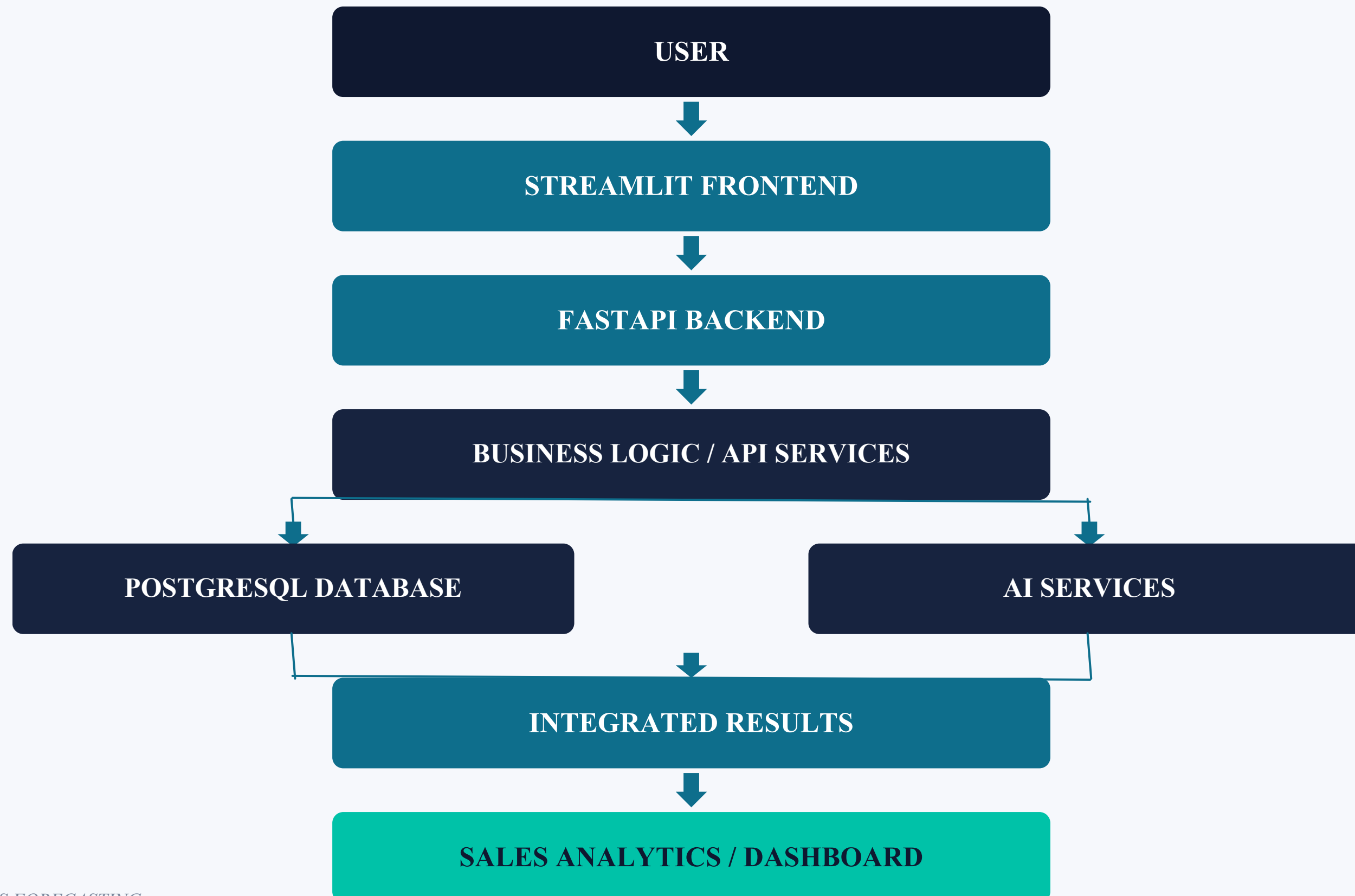


End-to-End Validation

Validate complete end-to-end functionality.

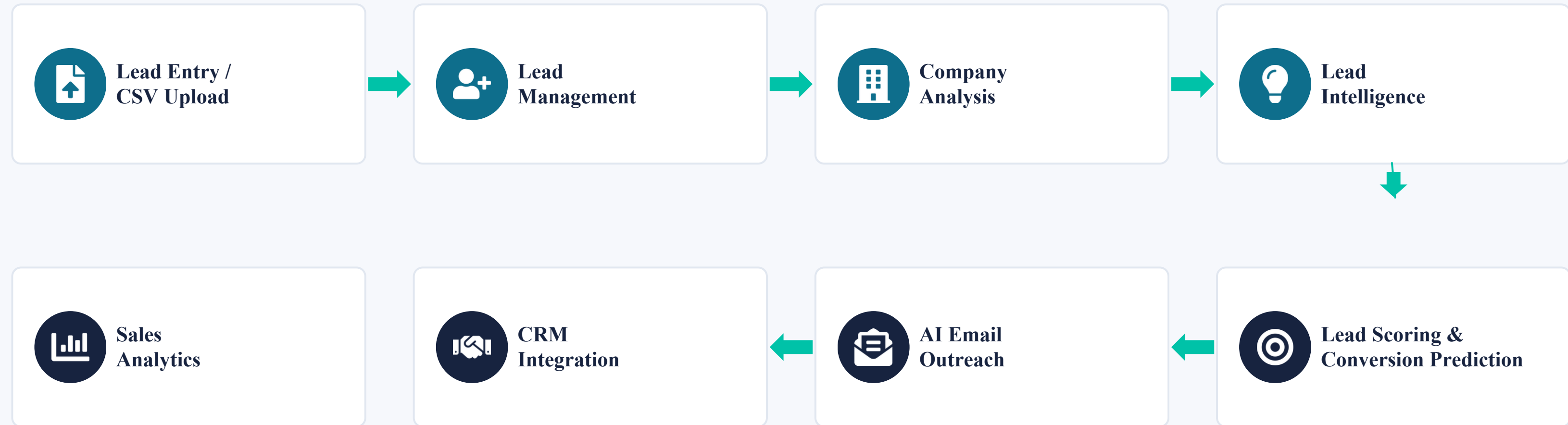


Integrated System Architecture





End-to-End Integration Workflow

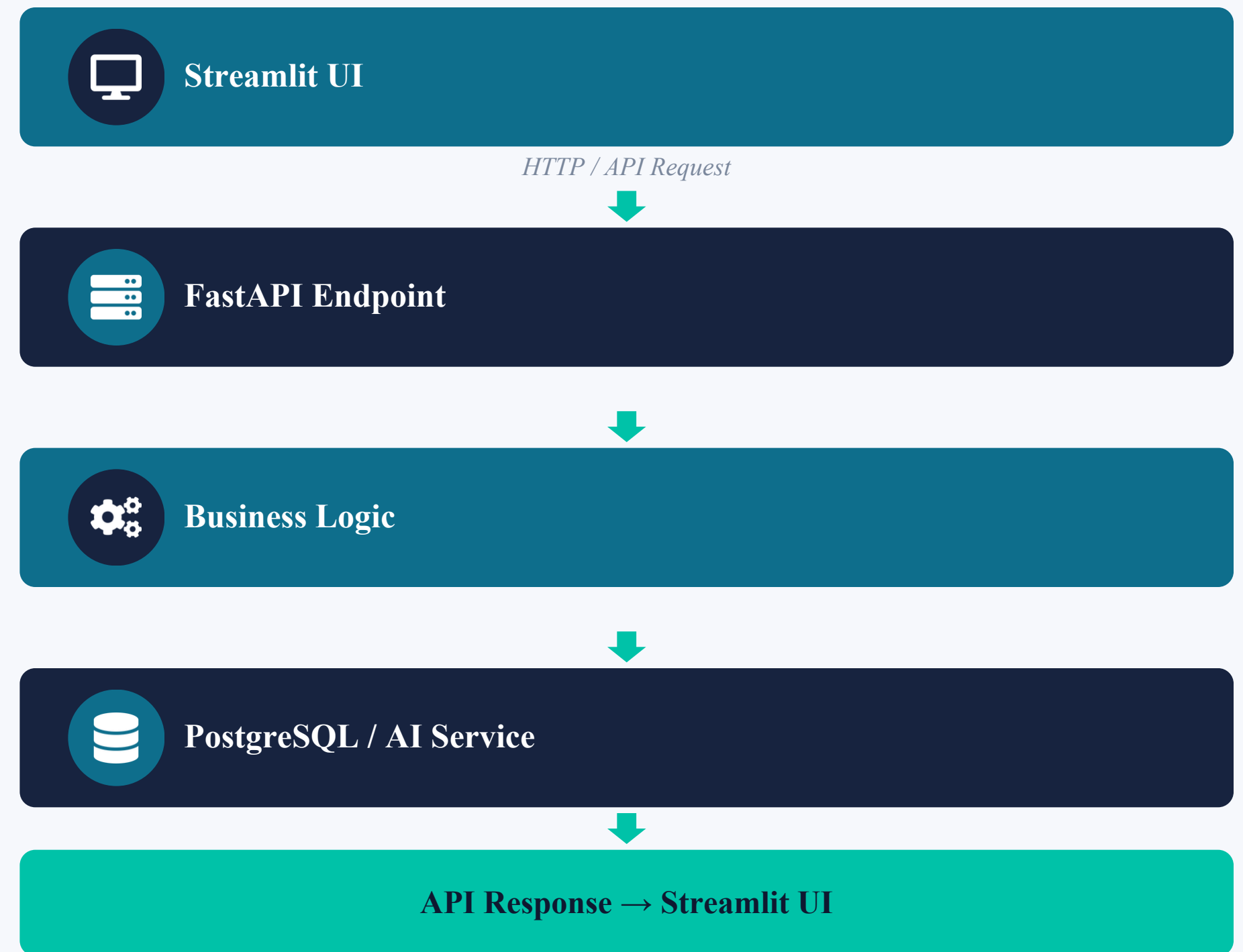


"Lead information flows through the integrated modules from acquisition and analysis to scoring, outreach, CRM synchronization, and sales insights."



Frontend-Backend-Database Integration

- Streamlit provides the interactive user interface.
- FastAPI provides REST API services for backend communication.
- Frontend requests are sent to the appropriate backend endpoints.
- Backend services process requests and perform business operations.
- PostgreSQL stores and retrieves lead and related business data.
- API responses are returned to the frontend and displayed to the user.





AI Services and CRM Integration



AI INTEGRATION

- Company information is used for AI-based analysis.
- AI generates lead intelligence and business insights.
- Lead attributes are used for lead scoring and conversion prediction.
- AI-powered email generation uses lead and company information.
- AI-generated results are integrated into the application workflow.



CRM INTEGRATION

- CRM communication is handled through FastAPI endpoints.
- Lead and contact information can be mapped to CRM structures.
- CRM synchronization supports centralized sales data management.
- Integration results can be tracked through synchronization logs.



Integration Testing & Validation

Test Scenario	Expected Result
Create Lead → Database	Lead stored successfully
Lead → Company Analysis	Analysis generated successfully
Lead → Lead Intelligence	Intelligence displayed successfully
Lead → Lead Scoring	Score and probability generated
Lead → AI Email	Personalized email generated
Frontend → Backend API	Correct response received
CRM Integration	Data synchronized successfully



VALIDATION APPROACH

- Verify data flow between connected modules.
- Validate API responses.
- Confirm database updates.
- Verify AI-generated outputs.
- Perform end-to-end workflow testing.



Module-wise Execution Time Analysis

Module / Operation	Execution Time	
Manual Lead Creation	< 5 sec	
CSV Lead Import	< 5 sec	
Company Analysis	< 5 sec	
Lead Intelligence	< 5 sec	
AI Email Generation	< 5 sec	
Lead Scoring & Conversion Prediction	< 5 sec	
CRM / API Integration	< 5 sec	



Integration Results & Conclusion

INTEGRATION RESULTS

- Frontend and backend successfully connected.
- Database operations integrated with backend services.
- AI services integrated into the lead workflow.
- Lead scoring and conversion prediction integrated.
- AI-powered email generation integrated.
- CRM / API integration established.
- End-to-end data flow validated.
- Execution time measured for major operations.



CONCLUSION

"Milestone 4 successfully integrates the major components of the AI-powered intelligent sales forecasting platform into a unified workflow. The integrated system connects the frontend, backend, database, AI services, and CRM layer while validating end-to-end functionality and measuring execution performance."